

GlowCart – Online Skincare Store Management System

ER Relationship Model

1. Customer Entity

Primary Key: Customer_ID

Attributes:

- Customer_ID (PK)
- First_Name
- Last_Name
- Email
- Phone_Number
- Password
- Gender
- Date_of_Birth
- Address
- City
- State
- Pincode
- Registration_Date

Relationships:

- A **Customer** places one or more **Orders**.
- A **Customer** owns one **Cart**.
- A **Customer** writes many **Reviews**.

2. Product Entity

Primary Key: Product_ID

Attributes:

- Product_ID (PK)
- Product_Name

- Description
- Price
- Stock_Quantity
- Skin_Type
- Ingredients
- Expiry_Date
- Brand_ID (FK)
- Category_ID (FK)

Relationships:

- A **Product** belongs to one **Brand**.
- A **Product** belongs to one **Category**.
- A **Product** appears in many **Cart Items**.
- A **Product** appears in many **Order Details**.
- A **Product** receives many **Reviews**.

3. Brand Entity

Primary Key: Brand_ID

Attributes:

- Brand_ID (PK)
- Brand_Name
- Country
- Website
- Contact_Email

Relationship:

- One **Brand** provides many **Products**.

4. Category Entity

Primary Key: Category_ID

Attributes:

- Category_ID (PK)
- Category_Name
- Description

Relationship:

- One **Category** contains many **Products**.

5. Cart Entity

Primary Key: Cart_ID

Attributes:

- Cart_ID (PK)
- Customer_ID (FK)
- Created_Date
- Total_Items

Relationships:

- One **Customer** owns one **Cart**.
- One **Cart** contains many **Cart Items**.

6. Cart_Item Entity

Primary Key: Cart_Item_ID

Attributes:

- Cart_Item_ID (PK)
- Cart_ID (FK)
- Product_ID (FK)
- Quantity
- Total_Price

Relationships:

- Each **Cart Item** belongs to one **Cart**.
- Each **Cart Item** refers to one **Product**.

7. Order Entity

Primary Key: Order_ID

Attributes:

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date
- Total_Amount
- Order_Status
- Shipping_Address

Relationships:

- A **Customer** places many **Orders**.
- One **Order** includes many **Order Details**.
- One **Order** has one **Payment**.
- One **Order** has one **Delivery**.

8. Order_Details Entity

Primary Key: Order_Detail_ID

Attributes:

- Order_Detail_ID (PK)
- Order_ID (FK)
- Product_ID (FK)
- Quantity
- Unit_Price
- Total_Price

Relationships:

- Each **Order Detail** belongs to one **Order**.
- Each **Order Detail** contains one **Product**.

9. Payment Entity

Primary Key: Payment_ID

Attributes:

- Payment_ID (PK)
- Order_ID (FK)
- Payment_Date
- Payment_Method
- Payment_Status
- Transaction_ID

Relationship:

- Each **Payment** is associated with one **Order**.

10. Delivery Entity

Primary Key: Delivery_ID

Attributes:

- Delivery_ID (PK)
- Order_ID (FK)
- Delivery_Address
- Courier_Name
- Tracking_Number
- Delivery_Status
- Delivery_Date

Relationship:

- Each **Delivery** is associated with one **Order**.

11. Review Entity

Primary Key: Review_ID

Attributes:

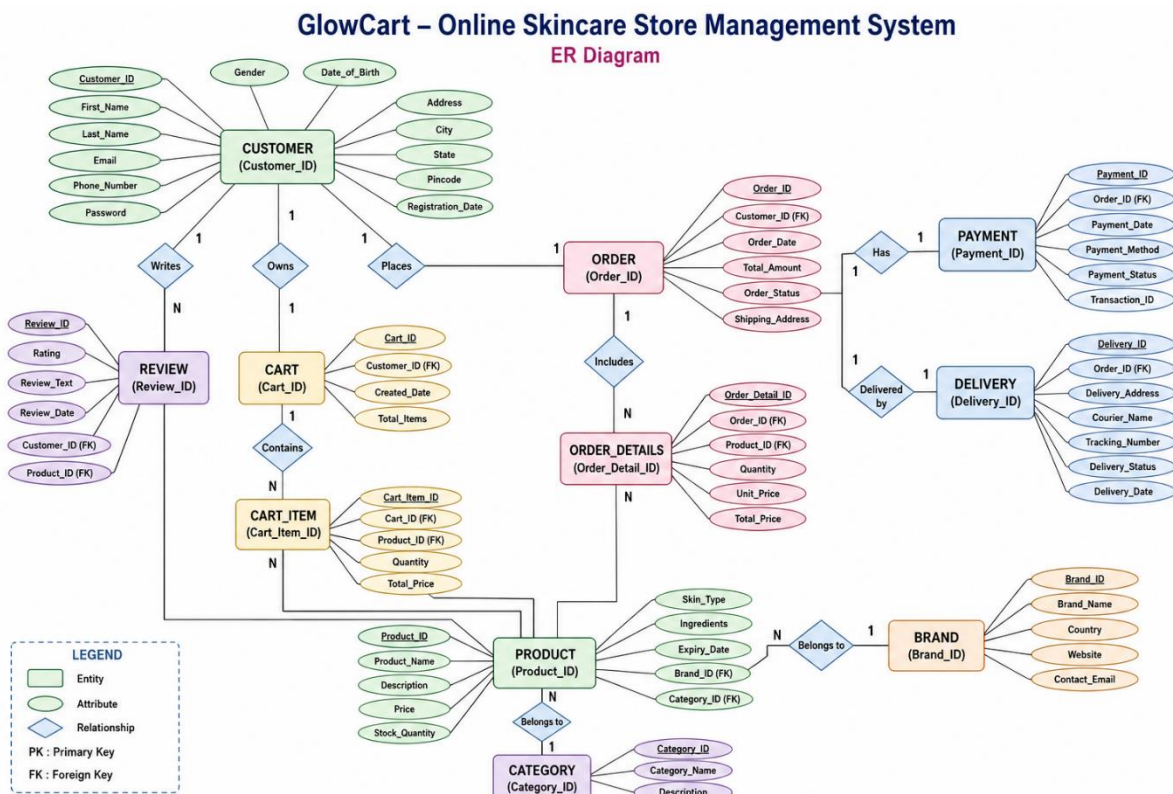
- Review_ID (PK)
- Customer_ID (FK)

- Product_ID (FK)
- Rating
- Review_Text
- Review_Date

Relationships:

- A **Customer** writes many **Reviews**.
- A **Product** receives many **Reviews**.

ER Model diagram:



Relationship Summary

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Order	1 : N

Entity 1	Relationship	Entity 2	Cardinality
Customer	Owns	Cart	1 : 1
Customer	Writes	Review	1 : N
Cart	Contains	Cart_Item	1 : N
Cart_Item	Refers to	Product	N : 1
Order	Includes	Order_Details	1 : N
Order_Details	Contains	Product	N : 1
Order	Has	Payment	1 : 1
Order	Has	Delivery	1 : 1
Product	Belongs to	Brand	N : 1
Brand	Provides	Product	1 : N
Product	Belongs to	Category	N : 1
Category	Contains	Product	1 : N
Product	Receives	Review	1 : N
Review	Written by	Customer	N : 1